

how much input things settle once dynamics settle

$$L[f'(t)] = sF(s) - f(0)$$

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

however, poles < 0

DC Gain: Transfer fct $G(0)$
System gain: $\frac{10(s+4)}{s+1}$

Fourier: What frequencies are present
Laplace: How system evolves over time

- Region of Convergence: Which exponential probes does this signal behave nicely.
- $\int_0^{\infty} f(t)e^{-st} dt = F(s)$ if e^{st} $\int_0^{\infty}$
- $a_n \frac{d^n x}{dt^n} + \dots + a_1 \frac{dx}{dt} + a_0 x = u(t)$ (LTI)
- $x(t) = \sum e^{(\sigma_i + j\omega_i)t} \cdot (\text{const})$

Poles, where system diverges.
If your system were spring, poles are the resonant frequencies where the spring wants to oscillate or yaw naturally

For rational $F(s) = \frac{N(s)}{D(s)}$ $s = \sigma + j\omega$

Cauchy's residue Theorem: $\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds = 2\pi j \sum \text{residues at poles} \cdot e^{st}$

So Inverse Laplace: $\frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds = f(t)$

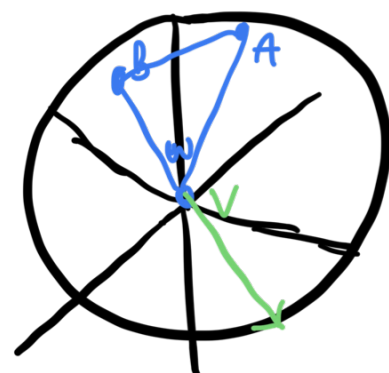
Root Locus: how poles of a system move in a complex plane as you vary some parameter - controller gain K

$$1 + KG(s) = 0 \quad \text{or} \quad D(s) + KN(s) = 0$$

Tracking modes of system as you adjust gain

If $G(s) = \frac{1}{s(s+2)}$ $1 + KG(s) = 0 \rightarrow s(s+2) + k = 0$
 $s = -1 \pm \sqrt{1-k}$

Quaternions:



$q = (w + \vec{v})$ is quaternion instead of using matrix for rotation, use quaternion.
Makes chaining easier too.

Transfer Functions

Output = (Something) * input

Transfer function: $G(s) = \frac{Y(s)}{U(s)}$

$Y(s)$ = Laplace of the output $y(t)$

$U(s)$ = Laplace transform of input $u(t)$

$G(s)$ = fct of s that tells you how the system responds to an input in Laplace Domain

Open-system: $Y(s) = G(s)U(s)$

Closed-system: $U(s) = C(s)(R(s) - Y(s))$
 $R(s)$ = Laplace of reference $C(s)$ = Controller

$\text{Res}[F(s), a] = \lim_{s \rightarrow a} (s - a) F(s)$

$T(s) = \frac{Y(s)}{U(s)} = \frac{G(s)C(s)}{1 + G(s)C(s)}$

PID: Proportional + Integral + Derivative

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

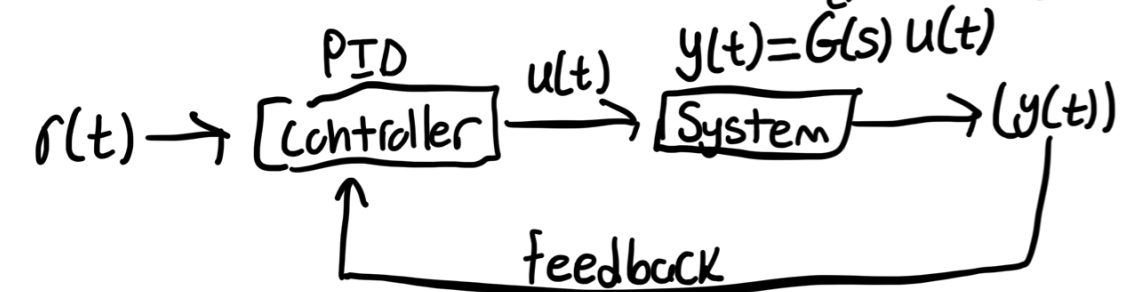
$$e(t) = r(t) - y(t)$$

$r(t)$ = reference

$y(t)$ = output

$u(t)$ = action

don't know, only know $y(t), r(t)$



Say you want $r(t)$, if sensor say $y(t)$ and $r(t) \neq y(t)$, throttle adjusts

LQR: State Space optimal controller for linear systems

$$\dot{x}(t) = Ax(t) + Bu(t)$$

- $x(t)$ = state vector
- $u(t)$ = control input
- A, B = system matrices

LQR chooses $u(t)$ to minimize a cost fct

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dx$$

if desired state $\neq 0$ $\tilde{x} = x(t) - x_{des}$ then $u(t) = -K\tilde{x}(t)$

Q = penalty on state deviations

R = penalty on control effort

How LQR produces a controller

$u(t) = -Kx(t)$: Solu gives state feedback

K = matrix gain computed by solving the Ricatti eq

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

P = Solu to Ricatti eq

$$K = R^{-1}B^T P$$

$$\dot{x} = (A - BK)x \quad (\text{closed-loop}) \rightarrow \text{Eigenvalues} = \text{closed loop poles}$$